

MUDITHA PRIYASAD

Automation | Mechatronics | Mechanical Engineer

IESL Associate Member (AM-32795)

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PROFESSIONAL SUMMARY

Automation and Mechanical Engineer with a BSc (Hons) in Mechanical Engineering, hands-on engineering exposure since 2019 and more than two years of industry delivery across 15+ projects. Experience in PLC-based control, HMI/SCADA support, sensors, pneumatics, stepper motion, commissioning, calibration, breakdown response and preventive/corrective maintenance. Delivered low-cost systems that raised fan-base filling throughput 6x and eyelet-punching output 4x. Brings practical fault isolation, technical documentation, TPM, operator guidance, cross-functional production support and digitization experience.

CORE SKILLS

- **Automation & PLC control:** Siemens STEP 7 Micro/WIN, Delta ISPSoft, Mitsubishi GX Works and Xinje PLC Software; ladder logic, sensors, actuators, pneumatics, stepper motion and PID concepts.
- **HMI/SCADA & fault diagnosis:** HMI troubleshooting, Haiwell SCADA improvement, relay/NPN/PNP input checks, expansion modules, three-phase control wiring and fault isolation.
- **Maintenance, reliability & commissioning:** Machine setting, adjustment, calibration, commissioning, preventive/corrective maintenance, breakdown response, root-cause analysis and recurring-fault reduction.
- **Documentation, safety & improvement:** Machine/process standards, SOPs, checklists, TPM, 5S, Kaizen, KPI tracking, safe-start controls, operator guidance and training support.
- **Engineering & digitization tools:** SOLIDWORKS, AutoCAD, MS Excel, Arduino IDE, Python, C++, Raspberry Pi, Marlin, GRBL, OrcaSlicer and Cura.

PROFESSIONAL EXPERIENCE

MARQO Industries | Mechanical Design & Automation Engineer

Dec 2024 - Present

- Established the company's engineering and automation function, including workshop systems, tooling control, technical documentation and supporting software workflows.
- Designed, commissioned, adjusted and improved production machines, jigs, fixtures and pneumatic mechanisms through requirements analysis, CAD, component selection, fabrication follow-up, testing and operator feedback.
- Supported production equipment through setup, breakdown response, mechanical, pneumatic, alignment and process fault isolation, followed by corrective improvements to reduce recurring interruptions.
- Delivered a fan-base filling machine (1 to 6 bases/min), a B22 eyelet puncher (3-4 to 16 pcs/min), a 1W LED-bulb assembly prototype (up to 20 pcs/min) and an ergonomic motor fixture.

MTone Designers | Founder

Jun 2021 - Present

- Delivered 20+ custom jobs, managing requirements, design, fabrication, inspection, costing and handover; developed MTone ERP to digitize job, quotation, inventory and costing workflows.
- Built, wired, calibrated and tested 3 custom FDM printers and 4 laser machines, integrating mechanics, motion systems, electronics, firmware and safe operating procedures.

MAS Active Shadowline | Engineering Intern

Apr 2024 - Nov 2024

- Supported TPM implementation through Fuguai tagging, autonomous-maintenance training, operational checks, machine/process standards, KPI tracking, 5S and Kaizen activities with production and technical teams.
- Prepared a coded machine-gauge reference and a phased six-section repair/service-centre layout covering inspection, part replacement, cleaning, reassembly, final checks, tooling, utilities and contractor coordination.
- Served as Interim Sustainability Executive alongside engineering duties, supporting practical resource-efficiency and workplace-improvement activities while maintaining engineering follow-up.
- Contributed to machine and process standards, repair-workshop layout improvements and the Align Tank automation initiative, focusing on safer handling, clearer workflows, maintainability and reduced downtime.

Effective Engineering & Automation | Automation Engineering Intern

Feb 2023 - Aug 2023

- Supported commissioning, preventive/corrective maintenance, breakdown response and field service for tensile-testing, concrete-mixer, oil-filling, glove-production and screen-printing equipment.
- Delivered a custom FDM printer end-to-end: mechanical architecture, procurement, wiring, BTT SKR 1.4/Marlin configuration, calibration, fault correction and print validation.
- Diagnosed Haiwell and Xinje PLC systems, checked relay/NPN/PNP inputs and expansion modules, tested ladder routines, completed three-phase control wiring and improved a Haiwell SCADA interface.

MAS Kreedha Vaanavil | Autonomation Engineering Intern

Jan 2019 - Apr 2019

- Integrated PLC sequencing, stepper motion, PID temperature control, SSR, thermocouple, heaters, sensors and pneumatic cylinders for a bonding-tape automation system.
- Supported iterative testing of temperature stability, positioning, cylinder sensing, fabric grip and two-hand start protection to improve repeatability, maintainability and operator safety.

SELECTED AUTOMATION & R&D PROJECTS

- **Fan-base filling machine:** Designed and commissioned a semi-automatic pneumatic system that increased output from 1 to 6 bases/min while reducing manual effort.
- **Custom FDM 3D printer:** Delivered mechanical architecture, procurement, wiring, BTT SKR 1.4/Marlin configuration, calibration, fault correction and print validation.
- **1W LED-bulb assembly machine:** Developed a semi-automatic prototype integrating sensors, pneumatics and control logic, with capacity up to 20 bulbs/min.
- **Bonding-tape automation machine:** Integrated PLC sequencing, stepper motion, PID heating, sensors, pneumatics and two-hand start protection.
- **MTone ERP software:** Digitized job tracking, quotations, inventory, material and machine costing and production follow-up in a maintainable desktop workflow.

INDUSTRIAL MACHINE & PROCESS EXPOSURE

- **Maintenance & reliability:** Preventive/corrective maintenance, breakdown response, commissioning, calibration, root-cause fault isolation, recurring-issue reduction and production support.
- **Automation & controls:** Siemens, Delta, Mitsubishi and Xinje PLC support; HMI/SCADA, sensors, pneumatics, stepper motion, PID concepts, three-phase control wiring and safe-start logic.
- **Documentation, safety & training:** Machine/process standards, SOPs, checklists, TPM, Fuguai tagging, autonomous-maintenance training, operator guidance and technical reports.
- **Manufacturing & digitization:** Cross-functional production support, KPI tracking, 5S, Kaizen, workshop planning, custom ERP workflows, rapid prototyping and engineering improvement.

EDUCATION

University of Jaffna | BSc (Hons) Mechanical Engineering

Oct 2017 - Feb 2024

- Final-year research: designed and fabricated the two-actuator bio-inspired hopping robot; led practical engineering projects and exhibition teams.

CERTIFICATIONS & TRAINING

- **SOLIDWORKS:** Design Professional (CSWP, 2026); CSWA - Mechanical Design; CSWA-AM - Additive Manufacturing.
- **Automation:** PLC Training - University of Moratuwa (2025); PLC Programming - Epic Engineering (2024).
- **Maintenance & controls:** Maintenance of Industrial Plant & Machinery (2026) and Industrial Motor Operation & Control (2025) - NERDC.
- **Manufacturing:** Advanced Injection Moulding Technology - Industrial Development Board of Ceylon (2026).

ACHIEVEMENTS & PROFESSIONAL ACTIVITIES

- Bronze Award / 3rd place for the University of Jaffna engineering-project display at Techno Sri Lanka 2023; participant at Air Tattoo Exhibition 2024 and SLT Robot Battles 2019.
- Organiser - University of Jaffna Battlegrounds 2020; IESL Associate Member AM-32795.

LANGUAGES

- English - Professional proficiency
- Sinhala - Native.